**DAY 1 AI ANSWERS AND MANUAL EXECUTION OUTPUT SCREENSHOTS**

# Linux File System Explorer

Description
**Investigation**

Ask AI:

1. **What is the purpose of /etc?** Stores system-wide configuration files that control how Linux and its services behave. Examples: /etc/hosts, /etc/passwd, /etc/ssh/ Used to configure networking, users, SSH, and other services.
2. **What is /var used for?** Stores data that changes frequently while the system runs. Examples: /var/log/, /var/cache/, /var/tmp/ Used for logs, caches, mail queues, and application data. Think of it as the system's changing records and activity folder.
3. **What is /home used for?** Stores personal files and settings for regular users. Example: /home/susmi/ Used for documents, downloads, projects, and user-specific configuration. Think of it as your personal workspace.
4. **What is /tmp used for?** Stores short-lived files created by applications and processes. Examples: temporary processing files and intermediate results. Files may be deleted automatically during reboot or by cleanup policies. Think of it as a scratchpad for temporary work.
5. **What is /usr used for?** Contains many installed applications, commands, libraries, and documentation. Despite its name, it is not primarily for personal user files. /usr/bin/ contains many everyday commands and programs. /usr/lib/ is libraries used by programs. /usr/share/ is shared data and documentation.

**Challenge question**

**If an application is generating log files, which Linux directory would you normally investigate first, and why?**

The answer is /var/log. By convention /var holds data that changes over time, and /var/log is the standard place for system and application logs




# Linux Commands Investigation

Description

**AI questions**

1. **What does pwd do?** pwd stands for Print Working Directory. It displays the full path of the directory you are currently in.
2. **Difference between ls and ls -la?** ls lists visible file names only. ls -la shows all files including hidden ones (-a) in long format (-l): permissions, owner, group, size, and modified date.
3. **What does cd do?** Changes the current directory. cd ~ goes to home directory, cd .. goes up one level, cd - returns to the previous directory, cd / moves to the root directory.
4. **What does mkdir do?** mkdir creates a new directory. mkdir -p a/b/c creates nested directories in one go.
5. **What does rm do?** rm removes (deletes) files. There is no recycle bin, so deletion is permanent. Use -r to remove directories recursively.

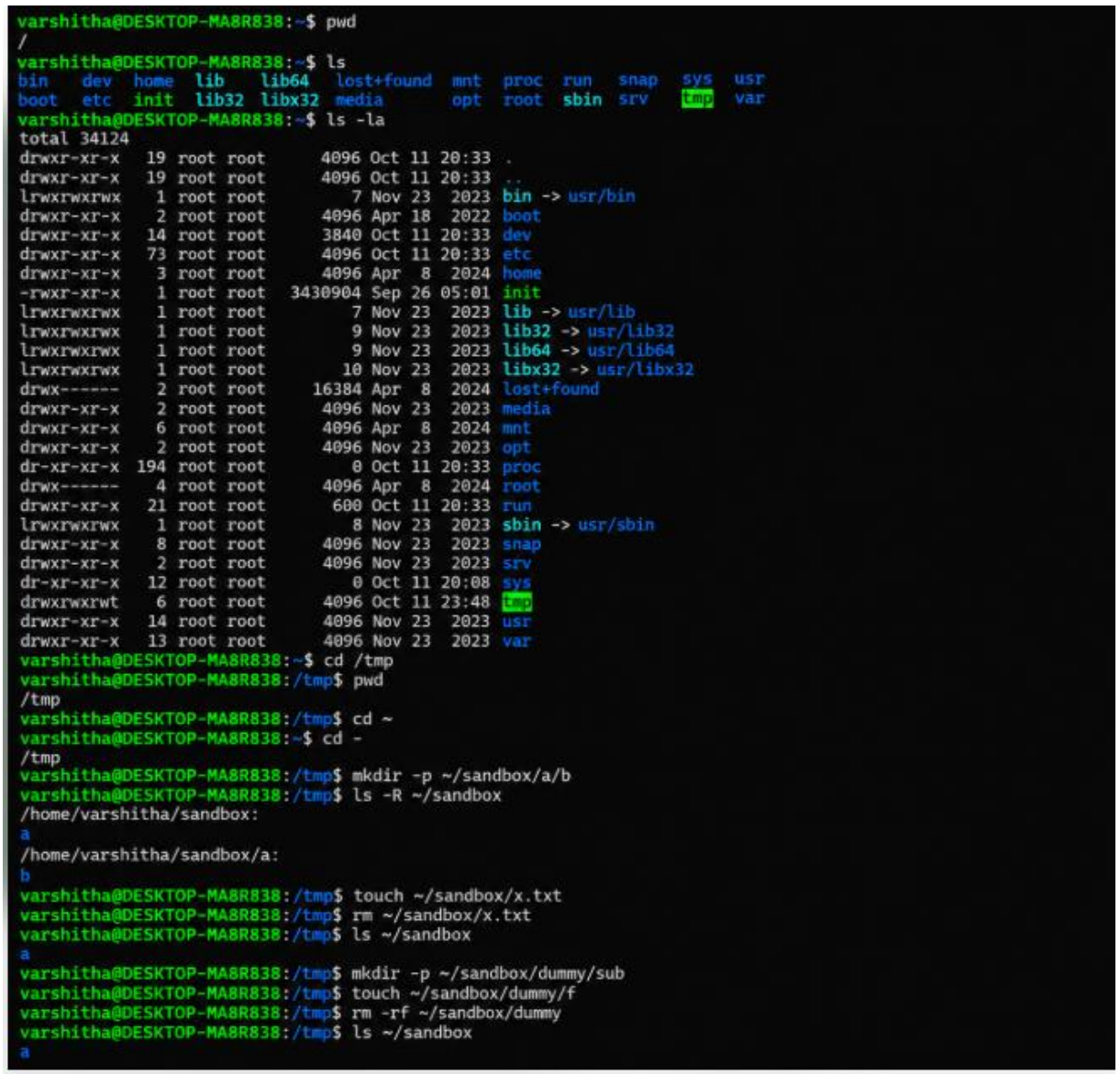
**Challenge**

Ask AI:

**"What happens if I execute rm -rf?"**

**Then discuss why this command is dangerous.** If I execute rm -rf, Linux deletes the specified files or folders and everything inside them without asking for confirmation in most cases. This command is dangerous because it can permanently delete important files, damage the operating

system, or cause data loss. Therefore, we should always check the file path carefully before using it and never run it on important system directories.



# Linux Permissions

Description

**AI questions**

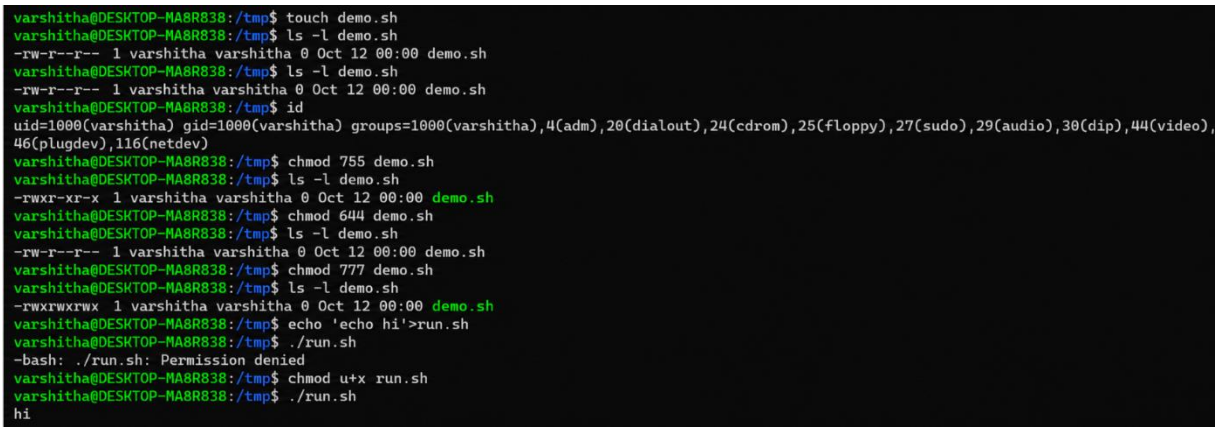
1. **What does rwx mean?** r = read (4), w = write (2), x = execute (1). For directories, x means you can enter/traverse it.
2. **What are owner, group and others?** Three permission classes: the file's owner (user), members of the file's group, and everyone else (others). In -rwxr-xr-- the three triplets are owner, group, others.
3. **What does chmod 755 mean?** Owner rwx (7), group r-x (5), others r-x (5). Common for scripts and directories: everyone can read and run, only the owner can modify.
4. **What does chmod 644 mean?** Owner rw- (6), group r-- (4), others r-- (4). Common for regular files: owner can edit, everyone else can only read.

5. **Why is chmod 777 generally discouraged?** It gives everyone read, write and execute. Any user or compromised process can modify or replace the file, which is a security risk, and it hides the real cause of the problem instead of fixing it.

**Challenge**

Ask:

**Is chmod 777 always required when a user gets "Permission denied"?** No, almost never. Check ls -l and the parent directories (x needed on each), identify who needs what access, then grant the minimum (chmod u+x, chmod 640, chown user:group). Other causes: wrong user, sudo needed, SELinux/AppArmor, or a read-only filesystem.




# Linux Processes

**Description**
This introduces a little troubleshooting thinking.

**AI questions**

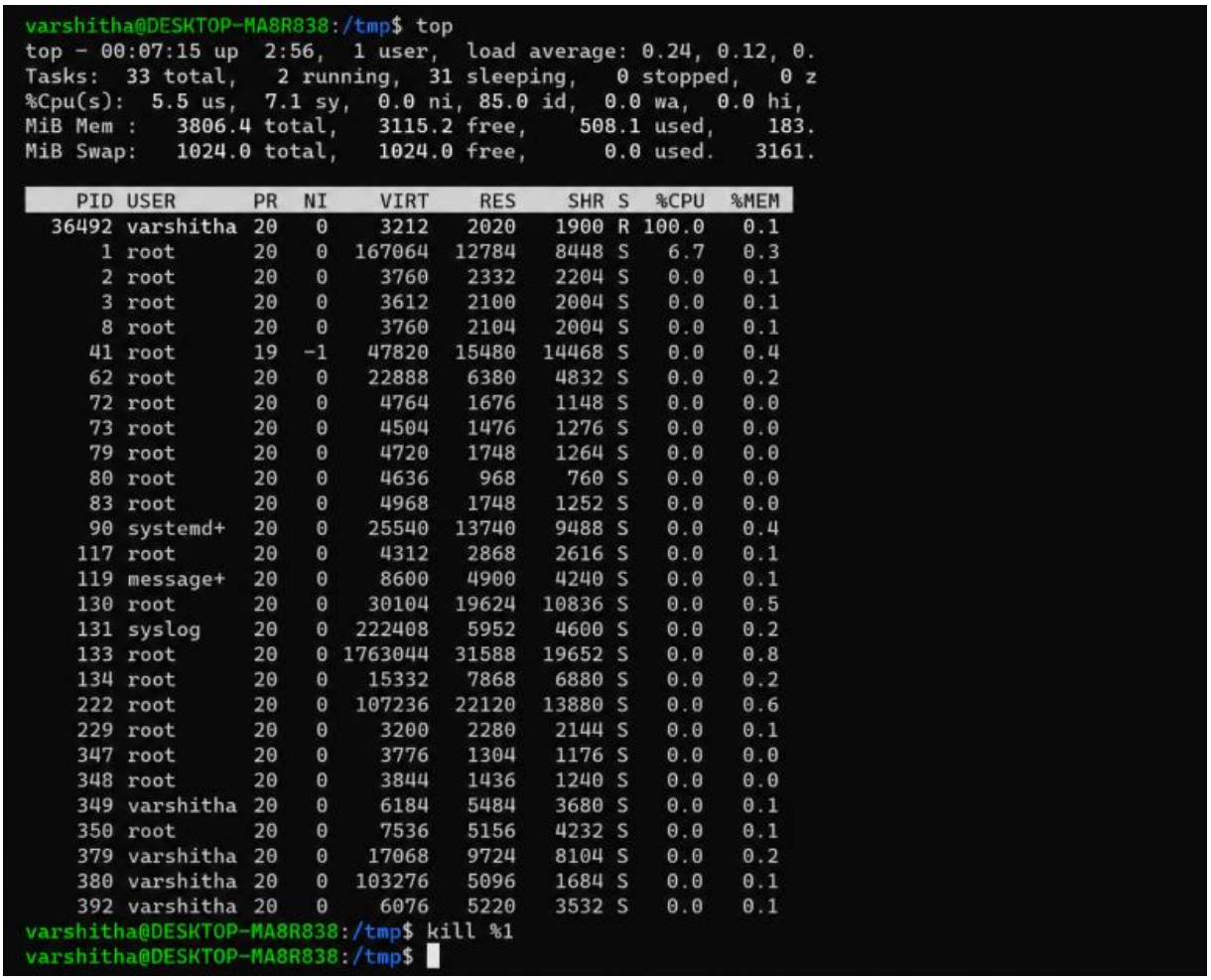
1. **What is a Linux process?**
2. **What does ps show?**
3. **What does ps aux show?**
4. **What does top do?**
5. **Why would a developer investigate processes?**

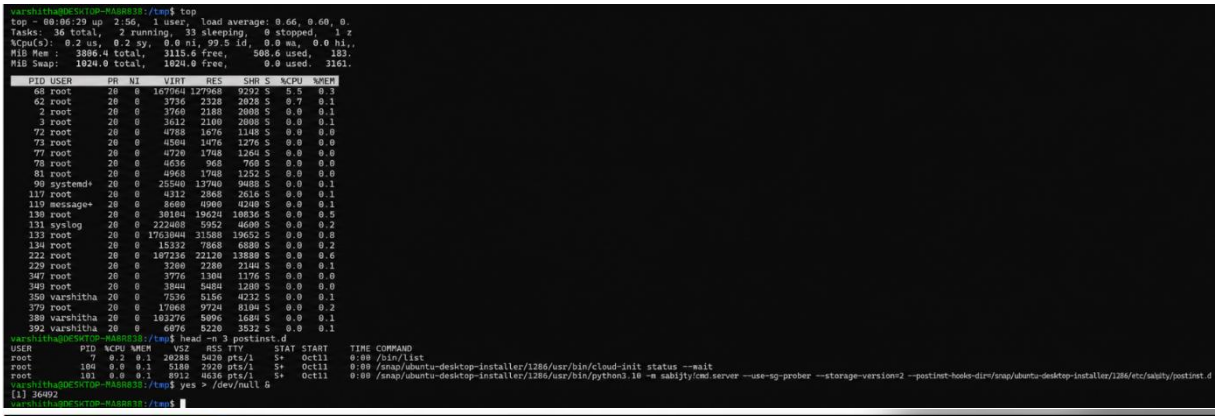
**Challenge**

Ask AI:

**"If an application is running slowly, can top help investigate the problem?"**

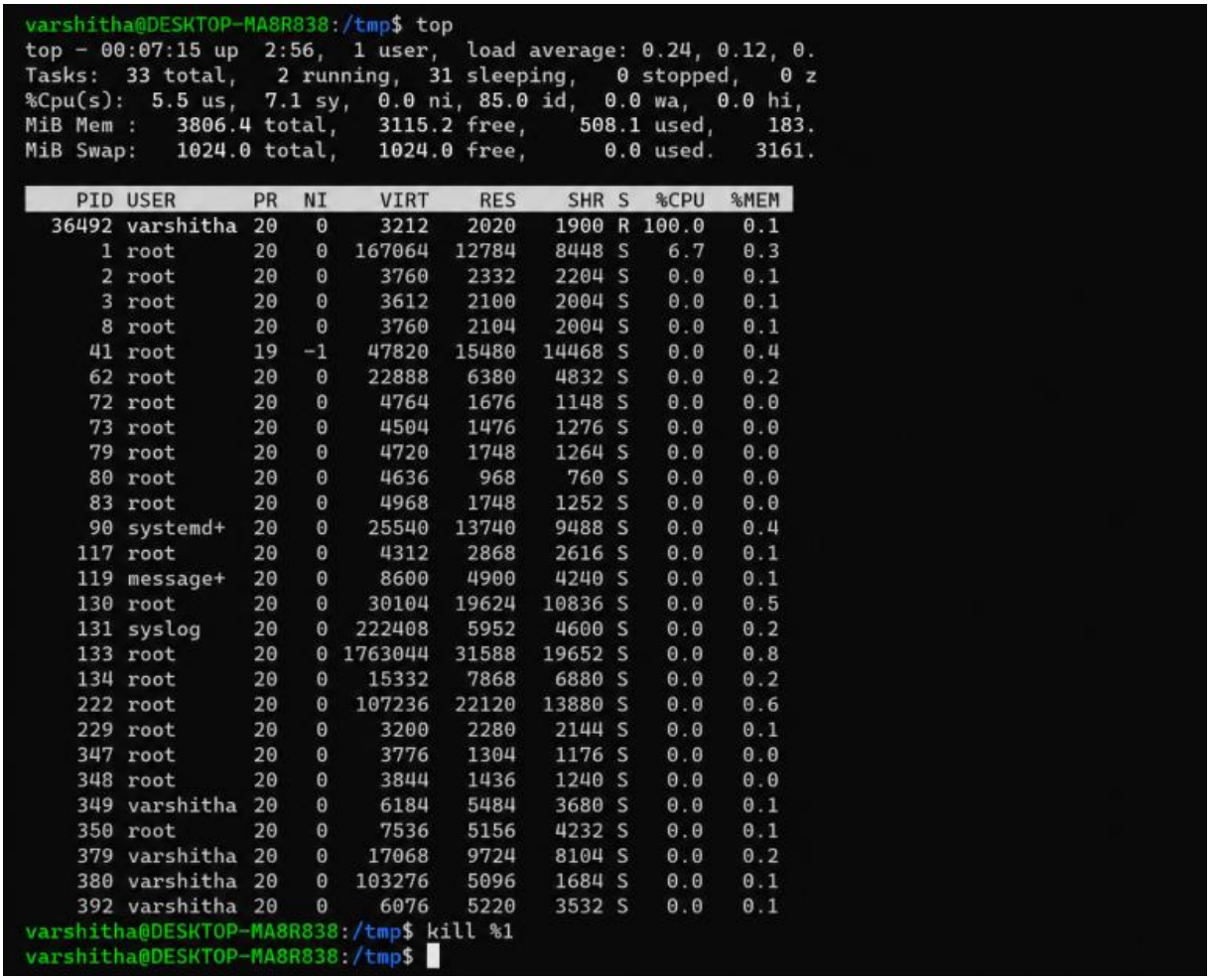
**Then explain what information they can actually obtain from it.** No, almost never. Check ls -l and the parent directories (x needed on each), identify who needs what access, then grant the minimum (chmod u+x, chmod 640, chown user:group). Other causes: wrong user, sudo needed, SELinux/AppArmor, or a read-only filesystem.


varshitha@DESKTOP-MA8R838:/tmp$ top
top - 00:07:15 up 2:56, 1 user, load average: 0.24, 0.12, 0.
Tasks: 33 total, 2 running, 31 sleeping, 0 stopped, 0 z
%Cpu(s): 5.5 us, 7.1 sy, 0.0 ni, 85.0 id, 0.0 wa, 0.0 hi,
MiB Mem : 3806.4 total, 3115.2 free, 508.1 used, 183.
MiB Swap: 1024.0 total, 1024.0 free, 0.0 used. 3161.
PID USER PR NI VIRT RES SHR S %CPU %MEM
36492 varshitha 20 0 3212 2020 1900 R 100.0 0.1
1 root 20 0 167064 12784 8448 S 6.7 0.3
2 root 20 0 3760 2332 2204 S 0.0 0.1
3 root 20 0 3612 2100 2004 S 0.0 0.1
8 root 20 0 3760 2104 2004 S 0.0 0.1
41 root 19 -1 47820 15480 14468 S 0.0 0.4
62 root 20 0 22888 6380 4832 S 0.0 0.2
72 root 20 0 4764 1676 1148 S 0.0 0.0
73 root 20 0 4504 1476 1276 S 0.0 0.0
79 root 20 0 4720 1748 1264 S 0.0 0.0
80 root 20 0 4636 968 760 S 0.0 0.0
83 root 20 0 4968 1748 1252 S 0.0 0.0
90 systemd+ 20 0 25540 13740 9488 S 0.0 0.4
117 root 20 0 4312 2868 2616 S 0.0 0.1
119 message+ 20 0 8600 4900 4240 S 0.0 0.1
130 root 20 0 30104 19624 10836 S 0.0 0.5
131 syslog 20 0 222408 5952 4600 S 0.0 0.2
133 root 20 0 1763044 31588 19652 S 0.0 0.8
134 root 20 0 15332 7868 6880 S 0.0 0.2
222 root 20 0 107236 22120 13880 S 0.0 0.6
229 root 20 0 3200 2280 2144 S 0.0 0.1
347 root 20 0 3776 1304 1176 S 0.0 0.0
348 root 20 0 3844 1436 1240 S 0.0 0.0
349 varshitha 20 0 6184 5484 3680 S 0.0 0.1
350 root 20 0 7536 5156 4232 S 0.0 0.1
379 varshitha 20 0 17068 9724 8104 S 0.0 0.2
380 varshitha 20 0 103276 5096 1684 S 0.0 0.1
392 varshitha 20 0 6076 5220 3532 S 0.0 0.1
varshitha@DESKTOP-MA8R838:/tmp$ kill %1
varshitha@DESKTOP-MA8R838:/tmp$




```
varshitha@DESKTOP-MA8R838:/tmp$ echo $$
349
varshitha@DESKTOP-MA8R838:/tmp$ ps -p $$
    PID TTY          TIME CMD
    349 pts/0    00:00:00 bash
varshitha@DESKTOP-MA8R838:/tmp$ ps
    PID TTY          TIME CMD
    349 pts/0    00:00:00 bash
  36610 pts/0    00:00:00 ps
varshitha@DESKTOP-MA8R838:/tmp$ ps aux | head
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START    TIME COMMAND
root           1  1.2  0.3 167064 127384 ?        Ss   Oct11   11:49 /sbin/init
root           2  0.0  0.0   3760   2332 ?        Sl   Oct11    0:00 /init
root           8  0.0  0.0   3612   2188 ?        S<   Oct11    0:00 /init-watch
root          41  0.0  0.3  47820  15480 ?        S<l  Oct11    0:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
root          62  0.0  0.0  22888   6380 ?        Ss   Oct11    0:00 /lib/systemd/systemd-journald
root          72  0.0  0.0   4784   6308 ?        Ss   Oct11    0:02 /lib/systemd/systemd-udevd
root          73  0.0  0.0   4564   1478 ?        Ss   Oct11    0:01 snapfuse /var/lib/snapd/snaps/core22_864.snap /snap/core22/864 -o ro,nodev,allow_other,suid
root          30  0.0  0.0   4720   1748 ?        Ss   Oct11    0:02 snapfuse /var/lib/snapd/snaps/bare_5.snap /snap/bare/5 -o ro,nodev,allow_other,suid
root          83  0.0  0.0   4968   1748 ?        Ss   Oct11    0:02 snapfuse /var/lib/snapd/snaps/snapd_20290.snap /snap/snapd/20290 -o ro,nodev,allow_other,suid
varshitha@DESKTOP-MA8R838:/tmp$ top
top - 00:06:07 up 2:55,  1 user, load average: 0.07, 0.08, 0.0
Tasks: 34 total,   1 running, 33 sleeping,   0 stopped,   0 z
%Cpu(s):  0.2 us,  0.2 sy,  0.0 ni, 99.5 id,  0.0 wa,  0.0 hi,
MiB Mem :   3806.4 total,    3116.7 free,     505.5 used,     184.
MiB Swap:   1024.0 total,    1024.0 free,       0.0 used.    3164.

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM
      1 root      20   0  167064 127384   8448 S   0.7   0.3
     68 root      20   0  943924  78552  22764 S   0.3   2.0
      2 root      20   0   37236  37568   9292 S   0.3   0.1
      3 root      20   0    3760   2332   2204 S   0.0   0.1
      8 root      20   0    3612   2100   2004 S   0.0   0.1
     41 root      19  -1   47820  15480  14468 S   0.0   0.4
     62 root      20   0   22888   6380   4832 S   0.0   0.2
     72 root      20   0    4764   1676   1148 S   0.0   0.0
     73 root      20   0    4504   1476   1276 S   0.0   0.0
     77 root      20   0    4720   1748   1264 S   0.0   0.0
     78 root      20   0    4636    968    760 S   0.0   0.0
     81 root      20   0    4968   1748   1252 S   0.0   0.0
     90 systemd+  20   0   25540  13740   9488 S   0.0   0.4
    117 root      20   0    4312   2868   2616 S   0.0   0.1
    119 message+  20   0    8600   4900   4240 S   0.0   0.1
    130 root      20   0   30104  19624  10836 S   0.0   0.5
    131 syslog    20   0  222408   5952   4600 S   0.0   0.2
    133 root      20   0 17630044  31588  19652 S   0.0   0.8
    380 varshitha 20   0  103276   5096   1684 S   0.0   0.1
    292 varshitha 20   0    6076   5220   3532 S   0.0   0.1
varshitha@DESKTOP-MA8R838:/tmp$ head -n 3 postinst.d
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START    TIME COMMAND
root           7  0.2  0.1  20288  5420 pts/1    S+   Oct11   0:00 /bin/bilist
root         104  0.0  0.1   5180  2920 pts/1    S+   Oct11   0:00 /snap/ubuntu-desktop-installer/1286/usr/bin/cloud-init status --wait
root         101  0.0  0.1   8912  4636 pts/1    S+   Oct11   0:00 /snap/ubuntu-desktop-installer/1286/usr/bin/python3.10 -m sabijty.cmd.server --use-sg-prober --storage-version=2 --postinst-hooks-dir=/snap/ubuntu/sabijty/postinst.d
varshitha@DESKTOP-MA8R838:/tmp$ yes > /dev/null &
[1] 36492
varshitha@DESKTOP-MA8R838:/tmp$
```


varshitha@DESKTOP-MA8R838:/tmp$ top
top - 00:07:15 up 2:56, 1 user, load average: 0.24, 0.12, 0.
Tasks: 33 total, 2 running, 31 sleeping, 0 stopped, 0 z
%Cpu(s): 5.5 us, 7.1 sy, 0.0 ni, 85.0 id, 0.0 wa, 0.0 hi,
MiB Mem : 3806.4 total, 3115.2 free, 508.1 used, 183.
MiB Swap: 1024.0 total, 1024.0 free, 0.0 used. 3161.
PID USER PR NI VIRT RES SHR S %CPU %MEM
36492 varshitha 20 0 3212 2020 1900 R 100.0 0.1
1 root 20 0 167064 12784 8448 S 6.7 0.3
2 root 20 0 3760 2332 2204 S 0.0 0.1
3 root 20 0 3612 2100 2004 S 0.0 0.1
8 root 20 0 3760 2104 2004 S 0.0 0.1
41 root 19 -1 47820 15480 14468 S 0.0 0.4
62 root 20 0 22888 6380 4832 S 0.0 0.2
72 root 20 0 4764 1676 1148 S 0.0 0.0
73 root 20 0 4504 1476 1276 S 0.0 0.0
79 root 20 0 4720 1748 1264 S 0.0 0.0
80 root 20 0 4636 968 760 S 0.0 0.0
83 root 20 0 4968 1748 1252 S 0.0 0.0
90 systemd+ 20 0 25540 13740 9488 S 0.0 0.4
117 root 20 0 4312 2868 2616 S 0.0 0.1
119 message+ 20 0 8600 4900 4240 S 0.0 0.1
130 root 20 0 30104 19624 10836 S 0.0 0.5
131 syslog 20 0 222408 5952 4600 S 0.0 0.2
133 root 20 0 1763044 31588 19652 S 0.0 0.8
134 root 20 0 15332 7868 6880 S 0.0 0.2
222 root 20 0 107236 22120 13880 S 0.0 0.6
229 root 20 0 3200 2280 2144 S 0.0 0.1
347 root 20 0 3776 1304 1176 S 0.0 0.0
348 root 20 0 3844 1436 1240 S 0.0 0.0
349 varshitha 20 0 6184 5484 3680 S 0.0 0.1
350 root 20 0 7536 5156 4232 S 0.0 0.1
379 varshitha 20 0 17068 9724 8104 S 0.0 0.2
380 varshitha 20 0 103276 5096 1684 S 0.0 0.1
392 varshitha 20 0 6076 5220 3532 S 0.0 0.1
varshitha@DESKTOP-MA8R838:/tmp$ kill %1
varshitha@DESKTOP-MA8R838:/tmp$